

PROFESSIONAL SUMMARY

Results-driven **Data Scientist** with **5+ years** of progressive experience architecting and deploying scalable, cloud-native data solutions on **AWS, Azure, and GCP**. Specializing in designing robust data pipelines, high-performance analytics platforms, and sophisticated GenAI applications leveraging advanced machine learning frameworks (**PyTorch, TensorFlow, Hugging Face**) and big data technologies (**PySpark, EMR, Airflow**). Proven ability to enhance business intelligence and optimize system performance through strategic DataOps practices, performance tuning, and cost-effective cloud implementations. Adept at integrating cutting-edge technologies such as **LLMs (GPT-4, LLaMA 2)**, **RAG pipelines (LangChain, Pinecone)**, and **distributed model training (DeepSpeed, FSDP)**, delivering measurable improvements in latency, accuracy, and operational efficiency. Passionate team leader with a track record of mentoring talent and driving high-impact projects that align technology closely with strategic business goals.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, R, JavaScript
- **ML Frameworks:** Pandas, Numpy, Matplotlib, Seaborn, scikit-learn, Keras, Django, Flask, PyTorch, TensorFlow, JAX (exploratory), Hugging Face Transformers
- **NLP & Gen AI:** spaCy, NLTK, AWS Comprehend, Hugging Face Transformers, BERT, GPT-3, T5
- **MLOps & Deployment:** Docker, Kubernetes, AWS SageMaker, Git, GitHub, Jenkins, CI/CD, AWS Lambda
- **Cloud/MLOps:** AWS (s3, Glue, EMR, Kinesis, Redshift, DMS, Step-functions, Cloud Formation, Lambda, EC2, SageMaker), Azure (Databricks, synapse, data factory), Google Cloud Platform (Big Query, Dataflow, Pub/sub)
- **Other:** Tableau, PowerBI, GitHub, Jenkins, Docker, Kubernetes, SAS Analytics, Jira, Sharepoint, Microsoft Office Suite

WORK EXPERIENCE

Client: Minutentag, New Jersey**Jan 2025 to Present****Role: Data Scientist - Analytics**

- Built a production-grade **image generation** pipeline with **Stable Diffusion, GPT-4, and ControlNet**, reducing generation latency by 40%.
- Building a modular **GenAI platform** where one **agent** enhances prompts (via GPT-4), another selects the **right image model (SDXL, ControlNet)**, and others handle input validation and feedback, like a small **AI production line**.
- Integrating memory, **vector search (Milvus/Pinecone)**, and tools for **real-time agent reasoning**.
- Developed human-in-the-loop feedback and **prompt engineering modules**, improving alignment accuracy by 92%.
- Containerized the **GenAI stack** and deployed **microservices** on **AWS SageMaker + EKS**.
- Designed automated **MLOps pipelines** with **Airflow** and **AWS Batch** for daily **retraining cycles**.
- Created **image filtering** and **classification models** to clean datasets for **multimodal tasks**.
- Led **social media integration** with Facebook, Instagram, and Twitter **APIs** to **automate content delivery**.

Client: Morgan Stanley, New York, NY**Aug'2023 to Dec'2024****Role: Senior Data Engineer**

- Developed deep learning models for loan default prediction using **TensorFlow, PyTorch, and AWS SageMaker**, improving prediction accuracy by 15%.
- Built **NLP models** for automating document classification (loan applications) using **spaCy** and **AWS Comprehend**
- Contributed to R&D of **RAG models** for financial advisory services, integrating **AWS Lambda** for real-time knowledge retrieval, improving recommendations by 10% with **Hugging Face Transformers**.
- Deployed models with **AWS SageMaker, Docker, and Kubernetes**, cutting deployment time by 25%
- Built end-to-end **ML pipelines** with **AWS Glue, AWS Lambda, and SQL**, optimizing **data ingestion** & reducing **model training time** by 20%.
- Trained **LLMs** for automating **financial report generation** and **customer interactions** using **AWS SageMaker**.
- Developed **fraud detection models - autoencoders** for **anomaly detection**, achieving an accuracy of 92%.
- Implemented real-time **data pipelines** for credit scoring using **AWS Data Pipeline** and **AWS Kinesis**.
- Developed automated **data cleaning** scripts for efficient preprocessing and **machine learning model** readiness.
- Applied validation rules, designed **automated scripts** in **SQL** and **Python** to ensure **data accuracy** and **detect outliers**.
- Designed scalable data workflows in **Databricks**, leveraging **Apache Spark** for high-performance data processing.
- Conducted **Exploratory Data Analysis (EDA)** using **Tableau, Matplotlib, and Seaborn** to uncover trends and insights.

Client: Vanguard, Charlotte, NC**Sep 2022 to July 2023****Role: Data Scientist**

- **Developed, trained, and optimized machine learning models** using **Scikit-learn, TensorFlow, and PyTorch**

- Performed **data preprocessing (cleaning, feature engineering & transformation)** with **Pandas, NumPy, and PySpark**
- Conducted **model evaluation** using **cross-validation, hyperparameter tuning**, and using **performance metrics**
- Containerized **ML models** using **Docker** and deployed them on **Kubernetes clusters** for **scalable & efficient** execution.
- Designed and implemented automated **data pipelines** for **feature extraction, transformation, and model training** using **AWS Lambda, Amazon SageMaker, MLFlow, AWS Glue, and Apache Airflow**.
- Developed & deployed **predictive analytics models** using **Logistic Regression, RandomForest, XGBoost, SVM**
- Implemented **CI/CD pipelines** for using **MLflow, GitHub Actions, and Jenkins**, for **seamless development & deployment**.
- Optimized model performance with feature selection (**Recursive Feature Elimination (RFE) & mutual information**).
- Utilized **ensemble learning methods** such as **Bagging, Boosting, and Stacking** to enhance **model robustness**
- Trained **time series forecasting** models using **ARIMA, Prophet, and LSTM** networks for **trend analysis & predictions**.
- Conducted **A/B testing** and **hypothesis testing** to evaluate **model effectiveness** and make **data-driven decisions**

Client: CTS, Bangalore, India

July 2020 to Aug 2022

Role: Data Analyst

- **Developed and optimized SQL data pipelines**, integrating multiple data sources and deploying predictive analytics models, resulting in a **12% improvement in reporting accuracy** and enhanced operational decision-making.
- **Designed and implemented interactive web interfaces** leveraging modern web technologies (**HTML5, CSS3, JavaScript, Bootstrap, SASS**) to visualize and present analytical insights dynamically.
- **Performed extensive data manipulation and exploratory data analysis (EDA)** using Python libraries including **Pandas, NumPy, Scikit-Learn, and Matplotlib**, enabling clear data-driven insights and accurate forecasting models.
- **Managed version control and deployment automation** with **GitLab, AWS CodePipeline, Jenkins, and AWS Elastic Beanstalk**, streamlining software development lifecycles and accelerating continuous integration and deployment (CI/CD).
- **Configured and deployed Apache Airflow DAGs** integrated with **AWS S3 and Snowflake** data warehouse, automating complex data workflows and significantly improving data pipeline efficiency.
- **Architected scalable data warehousing solutions and data processing pipelines**, enabling robust data management, analytics performance, and improved system scalability.

EDUCATION

Master of Science in Information Technology, Wilmington University, United States (3.7/4.0)

Bachelor of Technology, CVR College of Engineering, India (8.5/10.0)

PROJECTS

AI – Image Generator (LLM, Tensorflow, HuggingFace)

- Built a GenAI pipeline combining **Stable Diffusion XL, ControlNet, and LoRA** fine-tuning to generate lifestyle visuals.
- Designed a "Brain Layer" using **BLIP + GPT-4o** to analyze image, prompt, and select the best model for generation.
- Trained **LoRA adapters** on **Paperspace GPU** using images, dynamically integrating them into inference pipelines

RetailMLops – Price Prediction Pipeline (Google Cloud Platform, Regression)

- Utilized **airflow** for automated data pre-processing & model training pipeline, **CI/CD**, data & model version control.
- Enabled logging, model training & performance monitoring using **elk** along with real-time notifications via **slack**.

Twitter BOT Classification Pipeline (Sentiment Analysis, Text Classification, NLP)

- Developed a training pipeline, orchestrating model training, tracking multiple datasets & models via (**DVC, mlflow**)
- Experimented with multiple models & hyperparameters, logging & visualizing results with **MLflow** for model selection.

Intelligent Document Extraction & Search System (Azure Cognitive Search, OpenAI, Azure Document Intelligence)

- Developed an **Azure-native** pipeline integrating **DocuSign APIs, Azure Document Intelligence, and Azure Cognitive Search** to automate document extraction, chunking, and semantic search over large enterprise document stores.
- Enabled natural language Q&A over indexed documents using **Azure OpenAI**, improving document retrieval efficiency for business users.

AI-Powered Financial Report Assistant (AWS SageMaker, Pinecone, LangChain)

- Built a **RAG** pipeline combining **GPT models** with **Pinecone** vector search, deployed scalable endpoints on **AWS SageMaker + Lambda**, improving insight extraction speed by **40%**.

CERTIFICATIONS

- AWS Certified Data Engineer Associate
- AWS Certified Machine Learning Speciality
- AWS Certified Cloud Practitioner